Two streams in. A two-hour warning out.

Clinic chart

Age, labs, medicines
Blood pressure
Family history
Simulated gene flags

Wearable stream

Glucose every 15 min
Heart rate, steps, sleep
Past only
Nothing from later

Model

Patient-level split
Spike at 180 or above
Drop under 70
Peak sketch

Dashboard

Who should get a message?
Then one person
Green line so far
Orange estimate

Synthetic data only. Features stop at the current minute.

Two adverse events: glucose ≥180 (spike) and <70 (low).

Held-out ROC-AUC fusion 0.918 · glucose sensor only 0.828 · still under 140: fusion 0.873